Week 3 Recitation

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ECON8711 (Micro 1A)

Sep 6, 2024

1.1. True or false? If $V(y)$ is a convex set, then the associated production set Y must be convex.

$$Y \text{ convex } \underset{\not\leftarrow}{\overset{\circ}{\rightarrow}} V(y) \text{ convex}$$

Counter example ($\Leftarrow$):

$V(y) = \{x \in \mathbb{R}^2_+ \mid y \leq x_1^2 x_2^2\}$, $Y = \{(y, -x) \in \mathbb{R}^3_+ \mid y \leq x_1^2 x_2^2\}$

i) $V(y)$ is convex

Take any $x = (x_1, x_2)$, $x' = (x_1', x_2') \in V(y)$.

WLOG, we may assume $x \leq x'$.
(i.e., $x_1 \leq x_1'$ & $x_2 \leq x_2'$)

Fix $\lambda \in [0,1]$.

$\bar{x}_1 \equiv \lambda x_1 + (1-\lambda) x_1' \geq x_1$

$\bar{x}_2 \equiv \lambda x_2 + (1-\lambda) x_2' \geq x_2$.

As we know that $y \leq x_1^2 x_2^2$, $y \leq (x_1')^2 (x_2')^2$,

$$\bar{x}_1^2 \bar{x}_2^2 = (\lambda x_1 + (1-\lambda) x_1')^2 (\lambda x_2 + (1-\lambda) x_2')^2$$
$$\geq x_1^2 x_2^2 \geq y$$

So $(\bar{x}_1, \bar{x}_2) \in V(y)$.

Thus, $V(y)$ is convex.

ii) Y is not convex.

Choose $z = (1, -1, -1)$, $z' = (4, -1, 2)$, $\lambda = \frac{1}{2}$.

$z, z' \in Y$.

But $\lambda z + (1-\lambda) z' = \left(\frac{5}{2}, -1, -\frac{3}{2}\right) \notin Y$

as $\frac{5}{2} > (-1)^2 \left(-\frac{3}{2}\right)^2 = \frac{9}{4}$.

(σ)

1.2. What is the elasticity of substitution for the general CES technology $y = (a_1 x_1^\rho + a_2 x_2^\rho)^{1/\rho}$ when $a_1 \neq a_2$?

$$\sigma \equiv \frac{\frac{\Delta\left(\frac{x_2}{x_1}\right)}{\frac{x_2}{x_1}}}{\frac{\Delta TRS}{TRS}} = \frac{d \ln\left(\frac{x_2}{x_1}\right)}{d \ln TRS}$$

$$TRS \equiv -\frac{\frac{\partial f}{\partial x_1}}{\frac{\partial f}{\partial x_2}}$$

$$\frac{\partial f}{\partial x_1} = \frac{1}{\cancel{\rho}}(a_1 x_1^\rho + a_2 x_2^\rho)^{\frac{1}{\rho}-1} \cancel{\rho} a_1 x_1^{\rho-1}$$

$$= a_1 x_1^{\rho-1}(a_1 x_1^\rho + a_2 x_2^\rho)^{\frac{\rho-1}{\rho}}$$

$$\frac{\partial f}{\partial x_2} = a_2 x_2^{\rho-1}(a_1 x_1^\rho + a_2 x_2^\rho)^{\frac{\rho-1}{\rho}}.$$

$$\Rightarrow TRS = -\frac{a_1}{a_2}\left(\frac{x_2}{x_1}\right)^{1-\rho}$$

$$\Leftrightarrow \ln TRS = \ln\left|\frac{a_1}{a_2}\right| + (1-\rho)\ln\left(\frac{x_2}{x_1}\right)$$

$$\Leftrightarrow \underset{\sim\sim}{\frac{1}{1-\rho}} \ln TRS - \frac{1}{1-\rho}\ln\left|\frac{a_1}{a_2}\right| = \ln\left(\frac{x_2}{x_1}\right)$$

$$\sigma = \frac{d \ln\left(\frac{x_2}{x_1}\right)}{d \ln TRS} = \underset{\sim\sim}{\frac{1}{1-\rho}}$$

1.10. Let Y be a production set. We say that the technology is **additive** if $\mathbf{y}$ in Y and $\mathbf{y}'$ in Y implies that $\mathbf{y}+\mathbf{y}'$ is in Y. We say that the technology is **divisible** if $\mathbf{y}$ in Y and $0 \le t \le 1$ implies that $t\mathbf{y}$ is in Y. Show that if a technology is both additive and divisible, then Y must be convex and exhibit constant returns to scale.

pf) i) We shall show Y is convex.

Take $y, y' \in Y$. Fix $\lambda \in [0,1]$.

Due to the divisibility, $\lambda y \in Y$, $(1-\lambda)y' \in Y$.

By the additivity, $\lambda y + (1-\lambda)y' \in Y$

$\therefore$ Y is convex.

ii) Y exhibits CRS (i.e., $y \in Y \Rightarrow ty \in Y$ for $\forall t > 0$)

For $0 < t \le 1$, we can directly see Y exhibits CRS by the divisibility.

It only suffices to show only for $t > 1$.

Take $k > 1$,

$$k = \lfloor k \rfloor + \underbrace{(k - \lfloor k \rfloor)}_{\in [0,1]}.$$

the greatest integer $\le k$.

Then $ky = \lfloor k \rfloor y + (k - \lfloor k \rfloor) y$.

By the additivity, $\underbrace{y + y + \cdots + y}_{\lfloor k \rfloor - \text{times}} = \lfloor k \rfloor y \in Y$.

By the divisibility, $(k - \lfloor k \rfloor) y \in Y$.

Applying the additivity b/t $\lfloor k \rfloor y$ & $(k - \lfloor k \rfloor) y$, $ky \in Y$.

1.11. For each input requirement set determine if it is regular, monotonic, and/or convex. Assume that the parameters a and b and the output levels are strictly positive.

(a) $V(y) = \{x_1, x_2 \colon ax_1 \geq \log y, bx_2 \geq \log y\}$

(b) $V(y) = \{x_1, x_2 \colon ax_1 + bx_2 \geq y, x_1 > 0\}$

(c) $V(y) = \{x_1, x_2 \colon ax_1 + \sqrt{x_1 x_2} + bx_2 \geq y\}$

(d) $V(y) = \{x_1, x_2 \colon ax_1 + bx_2 \geq y\}$

(e) $V(y) = \{x_1, x_2 \colon x_1(1 - y) \geq a, x_2(1 - y) \geq b\}$

(f) $V(y) = \{x_1, x_2 \colon ax_1 - \sqrt{x_1 x_2} + bx_2 \geq y\}$

(g) $V(y) = \{x_1, x_2 \colon x_1 + \min(x_1, x_2) \geq 3y\}$

(SKIPPED)

9.1. **(a)** On page 203, I suggest that you look at the example $k = 2$ and $Z = \{(-1,1), (-3,2), (0,0)\}$. What is π in this example? What is the connection between this example and the discussion there?

False: If z^* is a subgradient of π at p, then $z^* \in Z^*(p)$, with $z^* \in Z$.

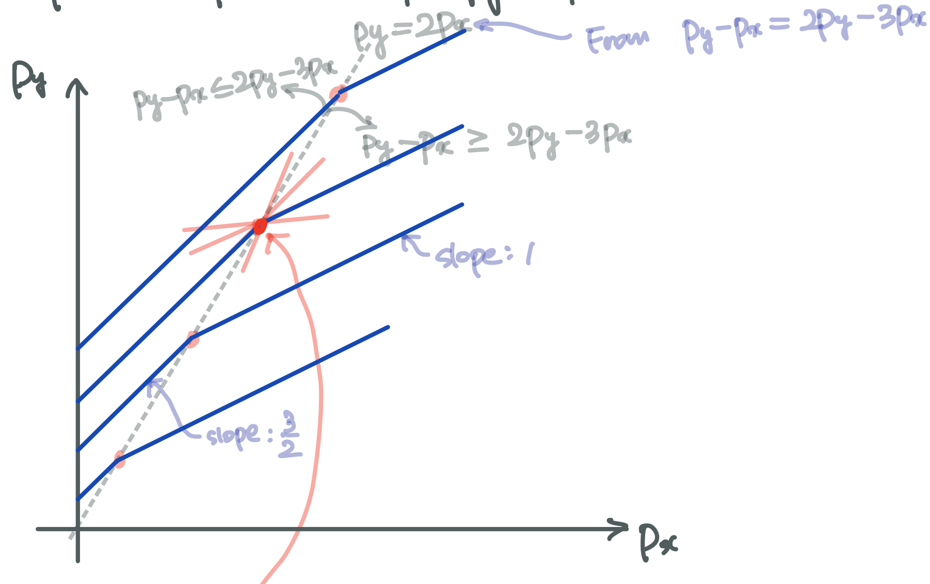
$$\pi(p) = \sup \{ p \cdot z \mid z \in Z \}.$$

Let $p \equiv (p_x, p_y) \in \mathbb{R}^2_{++}$

Then $\pi(p) = \max \{ p_y - p_x,\ 2p_y - 3p_x,\ 0 \}$

$\hookrightarrow$ Then
$$\begin{cases} (-1,1) \in Z^*(p) & \text{if } \pi(p) = p_y - p_x \\ (-3,2) \in Z^*(p) & \text{if } \pi(p) = 2p_y - 3p_x \\ (0,0) \in Z^*(p) & \text{if } \pi(p) = 0 \end{cases}$$

(Let's plot $\pi(p)$ on (p_x, p_y) plane!) — visualizing subgradients Z^*



we can find $z^* = (-z_1^*, z_2^*)$ s.t. $1 < -\frac{z_1^*}{z_2^*} < \frac{3}{2}$

For those z_1^*, $z^* \notin Z$.

(b) Provide the (simple) proof of Corollary 9.18. How does this connect with part (a) of this problem?

Corollary. *9.18. Suppose that a production possibility set Z that is closed and convex and has free disposal generates a profit function $\pi : \mathbb{R}^k_{++} \to \mathbb{R}$. Then if*

$Z = \overline{FDCH(Z)}$

$$\overline{FDCH(Z^*)} = Z^* := \{z \in \mathbb{R}^k | p \cdot z \leq \pi(p) \text{ for all } p \in \mathbb{R}^k_{++}\},$$

from Z

$Z^* = Z$. Moreover, $Z = Z^*$ is closure of the free-disposal convex hull of all the subgradients of π.

As $\pi(p)$ is generated from Z, we know that $Z \subseteq Z^*$.

We shall show $Z \supseteq Z^*$.

pf) By the way of contradiction (BWOC),

suppose $\exists\ \tilde{z} \in Z^*$ s.t. $\tilde{z} \notin Z$.

That is, for some $p \in \mathbb{R}^k_{++}$, $z^* \in Z$ s.t. $p \cdot z^* = \pi(p)$ (Z is closed)

$$p \cdot \tilde{z} \leq p \cdot z^* \quad \text{but} \quad \tilde{z} \notin Z.$$

$$\text{“} \sup_z \{p \cdot z \mid z \in Z\}.$$

Take $\bar{\varepsilon} \equiv p \cdot z^* - p \cdot \tilde{z} \geq 0$. By the definition of supremum,

$$\exists\ \bar{z} \in Z \text{ s.t. } p \cdot \bar{z} = \sup_z \{p \cdot z \mid z \in Z\} - \bar{\varepsilon}.$$

$$= p \cdot z^* - \bar{\varepsilon}$$

$$= p \cdot z^* - (p \cdot z^* - p \cdot \tilde{z})$$

$$= p \cdot \tilde{z}$$

$p \cdot (\bar{z} - \tilde{z}) = 0$, and as $p \in \mathbb{R}^k_{++}$,

$$\bar{z} - \tilde{z} = 0.$$

$\tilde{z} = \bar{z} \in Z$. (A contradiction).

9.5. Production possibility set Z is said to be *additive* if it has the property that if z and z' are both in Z, then so is $z + z'$. The idea is: If the firm can do z and can do z', then by setting up separate operations, it can do both z and z'.

(**a**) Show by example that additivity neither implies nor is implied by increasing returns to scale.

(**b**) Suppose Z is additive. What can you say about its profit function π?

* Additivity : rules out DRS

Here, " $z \in Z \Rightarrow tz \in Z$ for $\forall 0<t<1$. "

cf) IRS : $z \in Z \Rightarrow tz \in Z$ for $\forall t>1$.

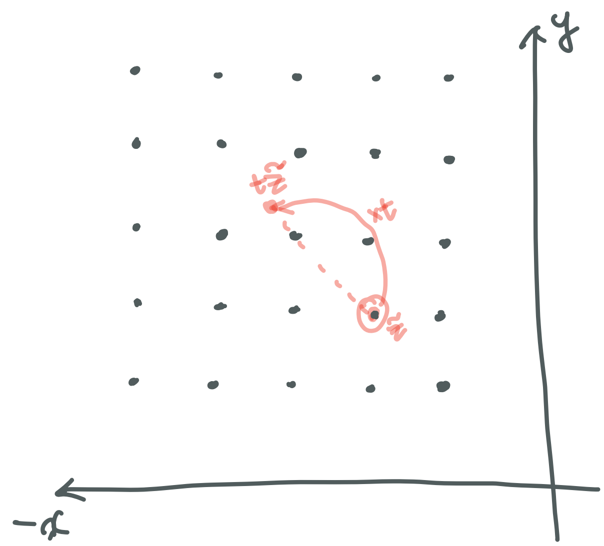
(a) $\text{Additive PPS} \underset{②}{\overset{①}{\not\rightleftarrows}} \text{IRS}$

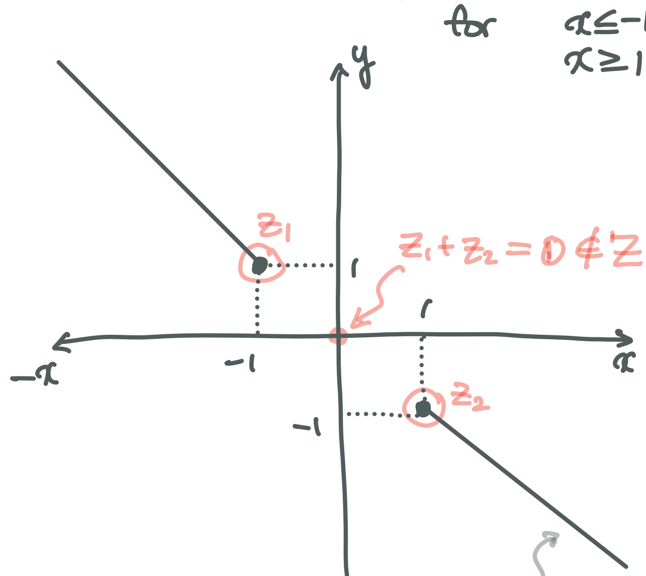
Counterexample

set of Integers

for ① : $Z = \mathbb{Z} \times \mathbb{Z}$

for ② : $Z = \{(-x, y) \in \mathbb{R}^2 \mid y = -x$ for $x \le -1$, $x \ge 1\}$





Reversible technology...(?)
Imagine it maybe...

(b) $p \cdot z$ is not bounded $(p \in \mathbb{R}^n_{++})$

$\Rightarrow \pi(p) = \sup_{z \in Z} \{p \cdot z \mid z \in Z\} = \infty$

unless $Z = \{0\}$. (In this case, $\pi(p) = 0$.)

Production Technology

About FDCH(Z)

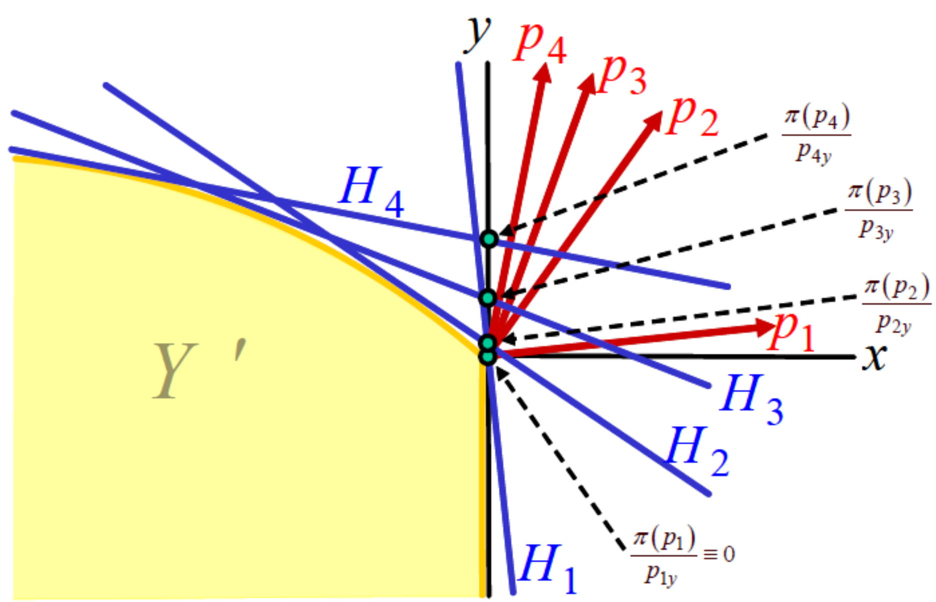
Definition (Dual Production Set)

The dual production set Z' of a production set Z is given by

$$Z' := \{z \in \mathbb{R}^k | p \cdot z \leq \pi(p) \text{ for all } p \in \mathbb{R}^k_{++}\}$$

where $\pi(p)$ is the profit function generated from Z.

- **The dual production set is bounded by the lower envelope of the hyperplanes defined by the various price vectors and points on the output axis as shown in the drawing below.**



Production Functions

Determining concavity of a function

CES Production: $f(x_1, x_2) = (a_1 x_1^\rho + a_2 x_2^\rho)^{\frac{1}{\rho}}$

Is this a concave function?

Determining Concavity

"Cookbook" procedure

Denote by

- $\pi = (\pi_1, \pi_2, \cdots, \pi_n)$: permutation of the integers $\{1, \cdots, n\}$
- A^π: the s symmetric $n \times n$ matrix applying the permutation π to rows and columns of A
- for $k \in \{1, \cdots, n\}$, A_k^π: $k \times k$ submatrix of A^π obtained by retaining only the first k rows and columns

Semi-definiteness

A **symmetric** $n \times n$ matrix A is

1. positive semidefinite iff $|A_k^\pi| \geq 0$ for all $k \in \{1, 2, \cdots, n\}$ and for all $\pi \in \Pi$.
2. negative semidefinite iff $(-1)^k |A_k^\pi| \geq 0$ for all $k \in \{1, 2, \cdots, n\}$ and for all $\pi \in \Pi$

Determining Concavity

"Cookbook" procedure

Concavity(Convexity) of a C^2 function

Let $f : \mathcal{D} \to \mathbb{R}$ be a C^2 function, where $\mathcal{D} \subseteq \mathbb{R}^n$ is open and convex. Then,

1. f is concave on $\mathcal{D}$ iff $D^2 f(x)$ is a negative semidefinite matrix for all $x \in \mathcal{D}$.
2. f is convex on $\mathcal{D}$ iff $D^2 f(x)$ is a positive semidefinite matrix for all $x \in \mathcal{D}$.
3. If $D^2 f(x)$ is a negative definite matrix for all $x \in \mathcal{D}$, then f is strictly concave on $\mathcal{D}$.
4. If $D^2 f(x)$ is a positive definite matrix for all $x \in \mathcal{D}$, then f is strictly convex on $\mathcal{D}$.

Determining Concavity

"Cookbook" procedure

CES Production: $f(x_1, x_2) = (a_1 x_1^\rho + a_2 x_2^\rho)^{\frac{1}{\rho}}$

1. Hessian of the function f?

2. Negative semidefiniteness of the bordered Hessian?